CYBERSECURITY PROJECT REPORT

VIRTUAL CYBERSECURITY LAB SETUP

WEEK 1 | CYBERSECURITY & ETHICAL HACKING | NETWORKWALKS

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Program / Batch	Cybersecurity & Ethical Hacking – Networkwalks (Batch B084)
Date	28 October 2026
Project Completed	Week 1 – Virtual Cybersecurity Lab Setup
Environment	Oracle VirtualBox with a Kali Linux virtual machine
Client / Target	None – personal practice lab on my own system
Permission Secured from Client?	Not applicable – no third-party systems were tested
Tools Used	Oracle VirtualBox, Kali Linux, NAT Network, Network Connections (NetworkManager)
Phases Covered	Lab preparation (before Phase 1: Reconnaissance & Footprinting)

1. Liability Disclaimer

I performed these activities only on my own computer and on virtual machines that I created and own. No third-party systems or networks were scanned, tested or accessed. All material in this report is for education and research purposes only. Do not use anything from it to break the law. The instructor, the authors and Networkwalks are not responsible for how this knowledge is used; every action you take is your own responsibility. Misuse can lead to criminal charges, heavy fines, loss of employment and a permanent record. In most countries, unauthorised access is a crime even when nothing is damaged.

2. Introduction

A safe, isolated lab is the foundation of every ethical hacking exercise. It lets a learner practise tools and techniques without putting real systems, data or networks at risk, and it can be reset quickly if something breaks.

This report covers my Week 1 project in the Cybersecurity & Ethical Hacking program at Networkwalks: building a virtual cybersecurity lab. I installed Oracle VirtualBox, imported Kali Linux as my attacking and testing machine, configured a custom NAT Network and the Kali IP settings, enabled clipboard sharing, drag-and-drop and a shared folder, confirmed network and internet connectivity, and saved a snapshot for recovery. This lab is the environment used for the later reconnaissance and scanning projects.

For each task below I describe the objective, what I did and why it matters. Screenshots taken from my screen recording are included in Section 8 as evidence.

3. Tools Used

The table below lists each tool or feature used in this project and its purpose.

Tool / Feature	Purpose
Oracle VirtualBox	Hypervisor used to create, run and manage the lab virtual machines, networks and snapshots.
Kali Linux	Penetration-testing distribution used as the attacking / testing machine.
NAT Network (VirtualBox)	Custom virtual network that gives lab machines a private subnet with outbound internet access.
Network Connections (NetworkManager)	Kali tool used to set the IP address, netmask, gateway and DNS server.
Shared Clipboard / Drag and Drop	VirtualBox settings used to move text and files between the host and the virtual machine.
Shared Folder	Folder shared between the host computer and the virtual machine for file transfer.
Snapshots	VirtualBox feature used to save the machine state and restore it later.

4. Activities Performed

4.1 Install and Configure Oracle VirtualBox

Objective: Set up a hypervisor on my computer to host the lab virtual machines.

What I did: I installed Oracle VirtualBox and used the VirtualBox Manager to manage my machines, snapshots and network settings.

Why it matters: Virtualisation keeps testing activity separate from my everyday operating system, so mistakes or malicious files stay inside the lab.

Evidence: Evidence 1.

4.2 Import and Configure Kali Linux

Objective: Add Kali Linux as the attacking / testing machine.

What I did: I imported Kali Linux into VirtualBox as a virtual machine named “kali (My fresh start …)”, started it, and logged in with the kali user account on the Kali login screen.

Why it matters: Kali Linux includes the reconnaissance, scanning and exploitation tools used throughout the program, so it is the main working machine for the lab.

Evidence: Evidence 1 and Evidence 2.

4.3 Configure a Custom NAT Network

Objective: Create a private virtual network for the lab with internet access.

What I did: I created a custom NAT Network in VirtualBox and attached the Kali virtual machine to it.

Why it matters: A NAT Network lets lab machines talk to each other and reach the internet while keeping them hidden from the physical network, which limits exposure.

4.4 Configure the Kali Linux IP Address

Objective: Give the Kali machine a fixed, predictable address on the lab network.

What I did:

- In Kali, I opened Network Connections and edited “Wired connection 1”.
- On the IPv4 Settings tab, I changed the method to Manual.
- I entered the address 10.0.0.2, netmask 24 (255.255.255.0) and gateway 10.0.0.1.
- I set the DNS server to 8.8.8.8 and saved the connection.

Why it matters: Static addressing makes it easier to identify lab machines and to document the network. The gateway lets the machine leave the local subnet, and the DNS server lets it resolve domain names.

Evidence: Evidence 3 and Evidence 4.

4.5 Configure Clipboard Sharing and Drag-and-Drop

Objective: Make it easy to move text and files between the host and the virtual machine.

What I did: I configured the virtual machine settings for clipboard sharing and drag-and-drop.

Why it matters: These features speed up lab work, but they also weaken the separation between host and guest, so they should be used with care (see Section 5).

4.6 Set Up a Shared Folder

Objective: Create a place to exchange files between the host computer and Kali.

What I did: I set up a shared folder between my host machine and the virtual machine.

Why it matters: A shared folder is useful for moving reports, wordlists and screenshots, but it is another path between the host and the lab.

4.7 Verify Network Connectivity and Internet Access

Objective: Confirm that the lab machine can use the network and reach the internet.

What I did: After saving the IP settings, I checked network connectivity and internet access from Kali, and recorded the result in the snapshot description: “i have set up the address and internet is working fine”.

Why it matters: Working connectivity and DNS are needed to update Kali, install tools and complete the reconnaissance exercises in the following weeks.

Evidence: Evidence 5.

4.8 Create a Virtual Machine Snapshot

Objective: Save a known-good state of the lab that I can return to.

What I did: I took a snapshot named “My fresh start on kali linux” on 28 October 2026 at 08:44 PM, after the network settings were working, and added a description of the working configuration.

Why it matters: A snapshot allows quick recovery if a tool, update or experiment breaks the machine, without rebuilding the lab from the start.

Evidence: Evidence 1 and Evidence 5.

5. Risk Analysis / Impact

The table below lists risks to consider when running a lab like this. They are general lab-security considerations, not vulnerabilities that I found or exploited.

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	Host and guest isolation weakened	Clipboard sharing, drag-and-drop and a shared folder were enabled.	Malicious files or data could pass from the virtual machine to the host computer.	● **Medium**
2	Default Kali account in use	The Kali image is logged in with the standard kali user account.	If the default password is not changed, anyone able to reach the machine could log in.	● **Medium**
3	Misconfigured network settings	The IP address, gateway and DNS were entered manually.	Wrong values cause loss of connectivity or place the machine on the wrong network.	● **Low**
4	External DNS resolver	DNS was set to 8.8.8.8, a public resolver.	Domain lookups from the lab are visible to a third-party DNS provider.	● **Low**
5	Sensitive data stored in snapshots	A snapshot preserves the full machine state.	Saved credentials, tool output or notes remain in the snapshot and can be restored or copied.	● **Low**

Risk level key: ● Medium ● Low

No exploitation or testing of any external system was performed in this project. The ratings are qualitative and are meant to guide safer lab practice.

6. Recommendations

Based on this project, I recommend the following practices for the lab:

1. **Keep the lab isolated**
   Use the NAT Network (or a host-only network) for lab machines and avoid bridging them to the physical network unless it is required.
2. **Change default credentials**
   Replace the default Kali password with a strong, unique one before using the machine for any real exercise.
3. **Limit host–guest sharing**
   Switch off clipboard sharing, drag-and-drop and shared folders when handling untrusted files or malware samples, or restrict them to one direction.
4. **Use snapshots before changes**
   Take a snapshot before installing tools, updating the system or running risky experiments, and delete old snapshots that hold sensitive data.
5. **Document the network plan**
   Keep a simple record of the subnet, gateway, DNS and the IP address of each lab machine.

6. **Keep software updated**
   Update VirtualBox and Kali Linux regularly so that known security issues are fixed.

7. **Test only with authorisation**
   Use the lab only against systems that I own or have written permission to test.

7. Conclusion

In Week 1 of my Cybersecurity & Ethical Hacking program, I built a working virtual cybersecurity lab. I installed Oracle VirtualBox, imported and configured Kali Linux, set up a custom NAT Network, configured a static IP address with a gateway and DNS server, enabled clipboard sharing, drag-and-drop and a shared folder, confirmed internet connectivity, and saved a snapshot as a recovery point.

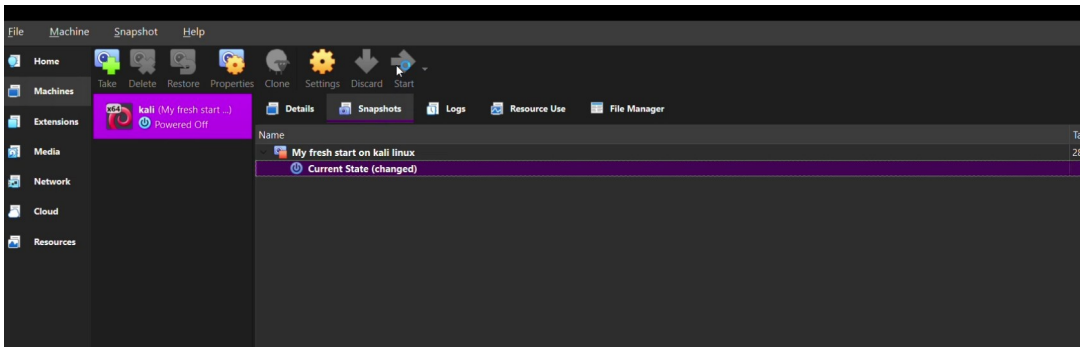
The project helped me understand virtual machine networking, NAT Network configuration, IP addressing, DNS resolution, shared folders and snapshots in a practical lab environment. I also learned that convenience features such as shared clipboards and folders need to be balanced against isolation, and that documenting each configuration step makes the lab easier to repeat and to troubleshoot.

This lab is the base for the following weeks, including the footprinting and reconnaissance exercises. All work was done on my own systems for educational purposes.

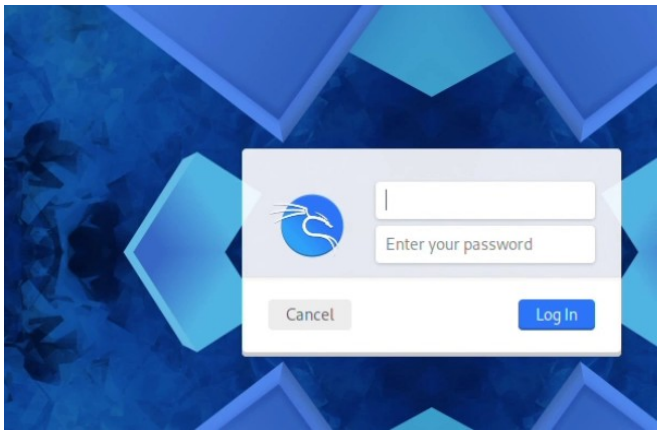
8. Evidences Collected

The following screenshots were taken from my screen recording of the lab and show the configuration described in Section 4.

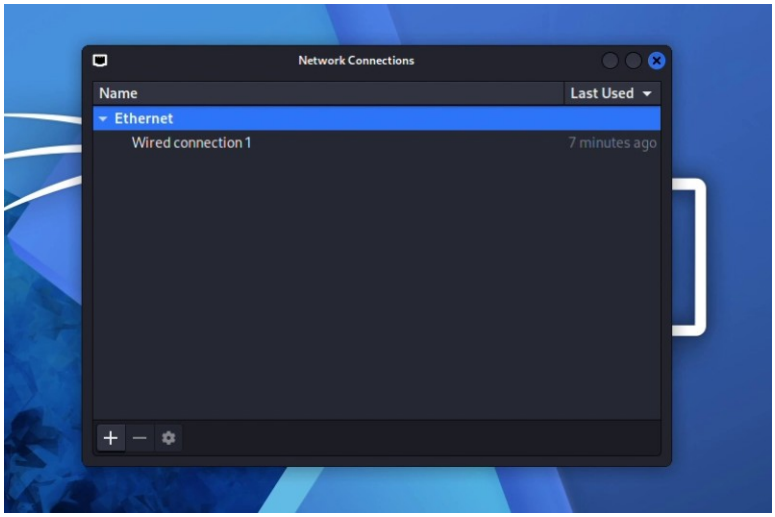
Evidence 1 – VirtualBox Manager showing the Kali Linux virtual machine and its snapshot



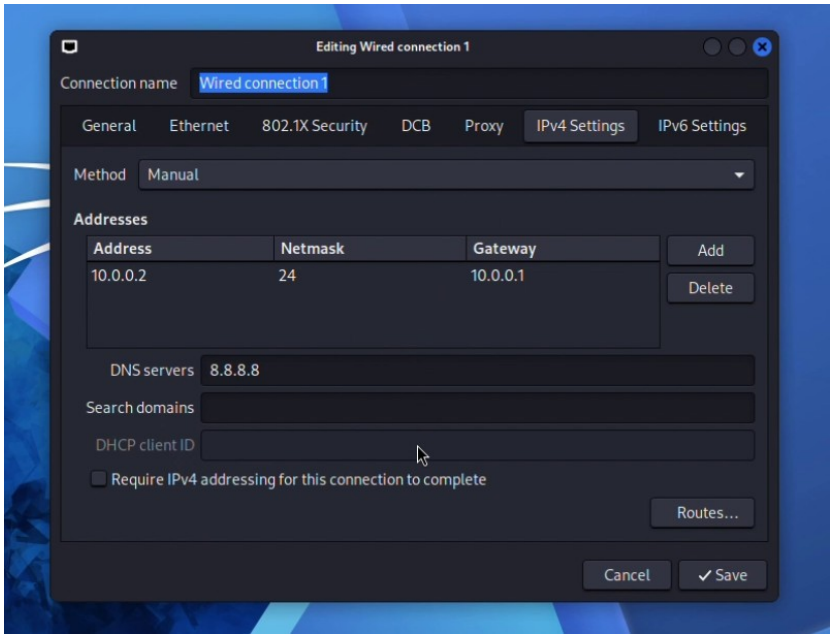
Evidence 2 – Kali Linux login screen after the virtual machine started



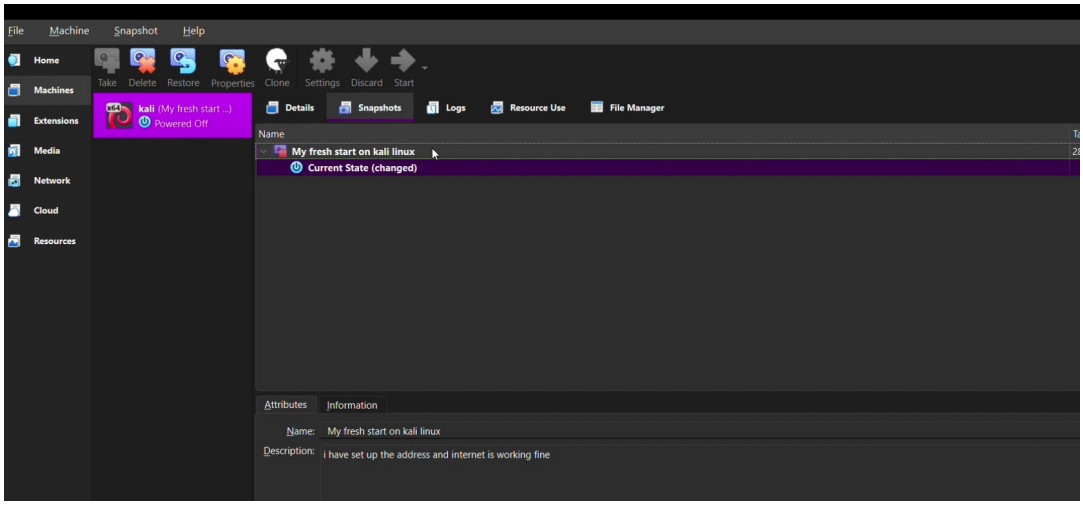
Evidence 3 – Network Connections window showing “Wired connection 1”



Evidence 4 – IPv4 settings: manual address 10.0.0.2/24, gateway 10.0.0.1, DNS 8.8.8.8



Evidence 5 – Snapshot “My fresh start on kali linux” with the description of the working configuration



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Project Information

Author	Shaik Mahboob Mohiuddin
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Week	Week 1
Project Title	Virtual Cybersecurity Lab Setup
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